Radiation - Lab Report

Szegedi Livia, Nguyen Ba Phi

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Abstract

This laboratory exercise introduces the fundamental principles of nuclear radiation detection through two complementary experiments. In the first section, a NaI(Tl) scintillation detector is assembled, calibrated, and characterized by examining the gain-voltage relationship of its photomultiplier tube, the signal time constants of the detector-electronics system, and the linearity of the spectroscopy amplifier. The energy spectrum of a Cs-137 gamma source is then recorded using a multichannel analyzer, and the influence of surrounding shielding materials on Compton backscattering is investigated. In the second section, a ^{3}He gas-filled proportional detector is used to measure neutron radiation from the training reactor, establishing its operating point and quantifying dead time losses at increasing reactor power levels.

Introduction

The detection of ionizing radiation relies on the charged particles produced when radiation interacts with matter. For gamma photons, three processes govern this interaction: the photoelectric effect, Compton scattering, and pair production. In a NaI(Tl) scintillation detector, the resulting electrons are converted into light flashes and then into electrical pulses by a photomultiplier tube, with pulse amplitude proportional to the absorbed photon energy. The first section of this exercise characterizes such a measurement chain by determining the PMT gain-voltage relationship, extracting the scintillator and electronic time constants from the detector pulse shape, verifying the linearity of the spectroscopy amplifier, and recording the Cs-137 energy spectrum under different shielding conditions to observe Compton backscattering effects.

The second section addresses neutron detection using a ^{3}He gas-filled proportional detector, which converts incident neutrons into detectable charged particles via the reaction $^3He + n \rightarrow p^+ + {}^3H^+ + 764$ keV. The tasks include establishing the optimal operating point of the detector by scanning the high-voltage and amplifier settings, determining the power supply to signal amplitude characteristics, setting up the single-channel analyzer at its operating point, and

measuring the neutron count rate at increasing reactor power levels to observe and quantify the dead time effect.

Section 1: Measurements with a scintillation detector

Task 1: Analysis of the amplitude of the generated signal

For the first section of the laboratory, the NaI(Tl) scintillation detector was placed into the lead tower together with the Cs-137 gamma source. The detector was connected to the high-voltage (HV) power supply via the NIM HV adapter, and the raw anode signal was monitored on an Agilent DSO-X 2002A digital oscilloscope. The signal output was also fed into the Canberra NIM 2022 spectroscopy amplifier, whose output was connected to the second oscilloscope channel, allowing both the raw detector pulse and the shaped amplifier output to be observed simultaneously. The full instrumentation chain — HV power supply, amplifier, multichannel analyzer (MCA), and single-channel analyzer (SCA) — was housed in a NIM bin frame, as shown in Image 1a below.

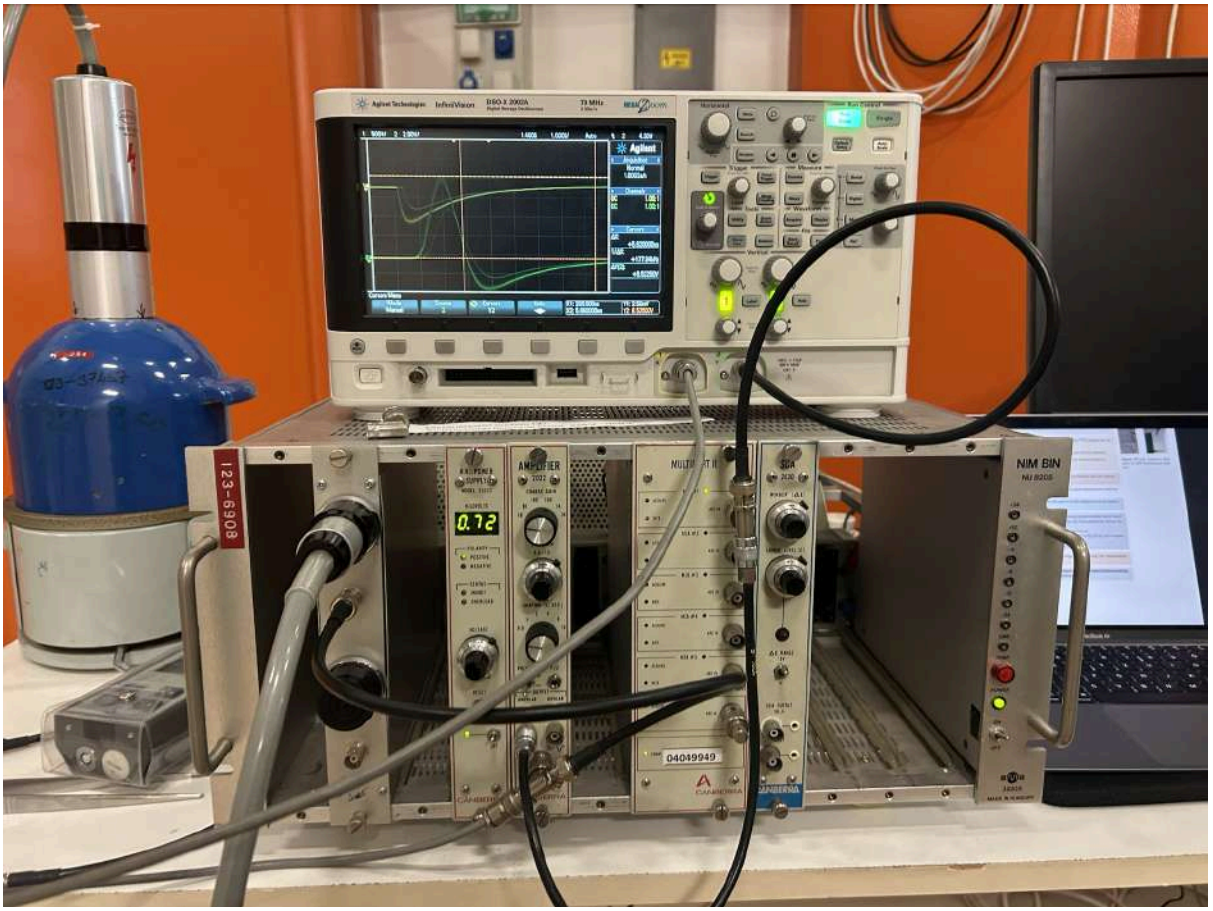


Image 1a: An image of the setup with the oscillator tube on the left and the blue lead dome storing the cesium. The oscilloscope on top. Below it is the HV Power supply, amplifier, multichannel analyzer (MCA) and single channel analyzer (SCA) from left to right.

Task 1. Determine the amplitude of the signal generated by the detector (V_{sig}) as a function of the supply voltage (V_{sup}) by varying the supply voltage (in $50V$ increments) between the ranges provided on the information sheet at the measurement station.

In this task, we aimed to determine the relationship between the detector's signal amplitude Vsig and the supply voltage Vsup. To do this, we placed the Cs-137 gamma-ray source inside the lead tower along with the scintillation detector. We then systematically varied the supply voltage in 50V increments from 450V up to 950V, recording the maximum signal amplitude on the oscilloscope at each step.



Image 1b, 1c: Image of the oscilloscope showing oscilloscope signal detecting a neutron (yellow line) and the processing (green line). 1b shows the amplitude signal when Vsup is 450V resulting in 21.125mV Vsig, 1c shows the amplitude signal when Vsup is 850V, resulting in 3.28V Vsig.

The raw signals observed on the oscilloscope (yellow trace) appear as negative pulses (upside down) because the output of the photomultiplier tube (PMT) is collected from the anode, which collects electrons and results in a negative voltage drop. The green trace, which represents the processed signal from the amplifier, appears cut off at the top and bottom because the amplifier saturates and limits the signal when the voltage exceeds its operational range (around 10V), leading to a clipped waveform.

Hint for the Lab Report. Plot the measured data points and fit $V_{sig} = a \cdot V_{sup}^{b}$ function to the measured points in order to determine the kind of power function that describes the dependence between the amplification of the PMT and the supply voltage (see Eq. II.1.2).

To analyze the behavior of the photomultiplier tube, we plotted the measured absolute signal amplitudes against the corresponding supply voltages. We then performed a curve fit using the power law function **Vsig = a * Vsup ^b** to empirically determine the exponent b, which characterizes the multiplication factor of the PMT.

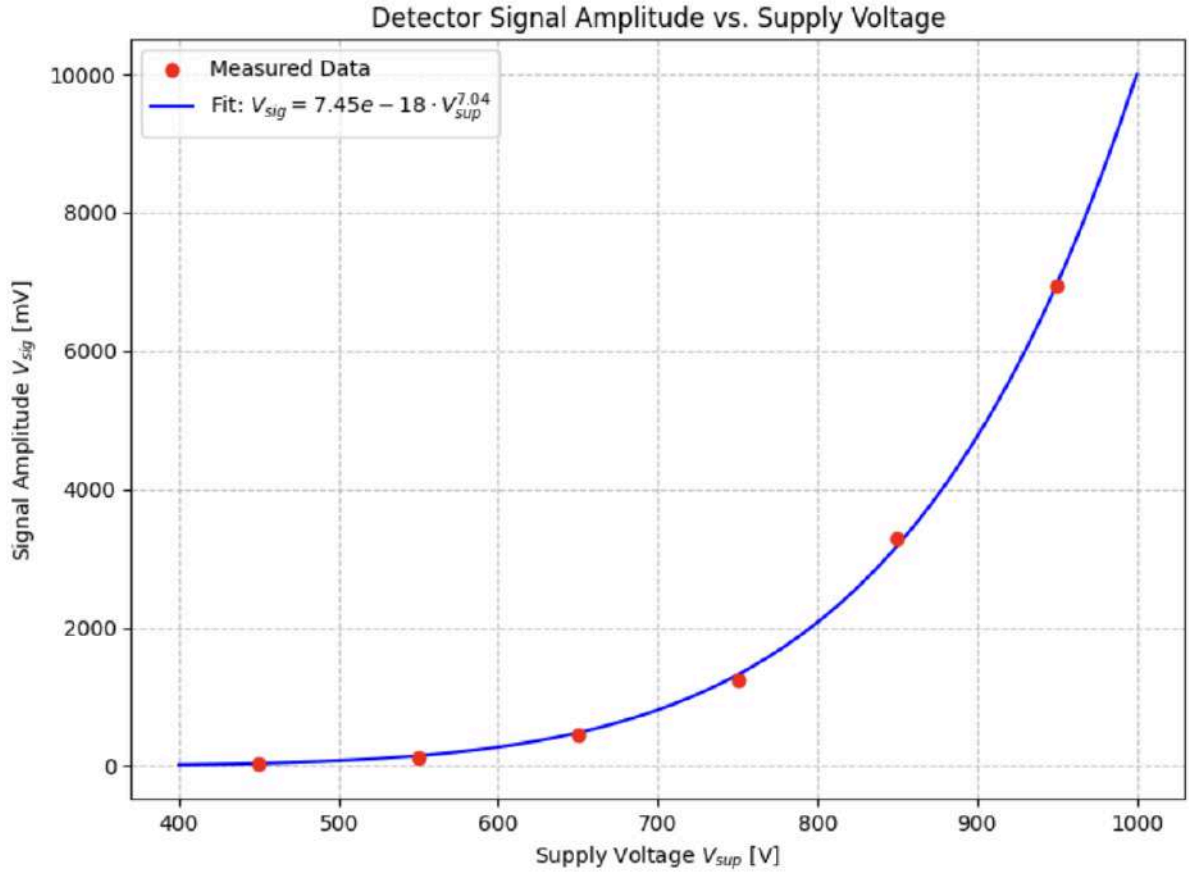


Figure 1. Measured data points and fitted Vsig = a * Vsup ^b

Based on the execution, the fitting process successfully found the parameters:

a = 7.45e-18 b = 7.04

Note: The calculated exponent (b = 7.04) aligns perfectly with the theoretical expectations detailed in your lab manual (Eq. II.1.2, page 12), which states that the fractional exponent for the multiplication factor of a photomultiplier tube is typically between 6 and 9.

Task 2: Exponential time constant t and time constant RC

> **Task 2.** Determine the exponential time constant τ of the scintillator crystal and the time constant RC of the electronics.

For the second task, our objective was to determine the exponential time constant of the scintillator crystal, τ, and the electronic time constant, RC. We adjusted the high-voltage supply to 670 V so that the absolute amplitude of the detector signal for the full-energy peak reached approximately 500 mV.



Image 2. A close-up of the signal wave at 670V applied voltage to form a 500mV signal amplitude

> **Hint for the Lab Report.** Fit a function corresponding to Eq. II.1.5 to determine the RC and τ time constants. Since Q is also an unknown parameter, it is sufficient (and

recommended) to combine the $\frac{RC\tau}{RC-\tau}\frac{Q}{\tau C}$ term into a single variable, when performing the fitting.
Compare the obtained values with the literature data!

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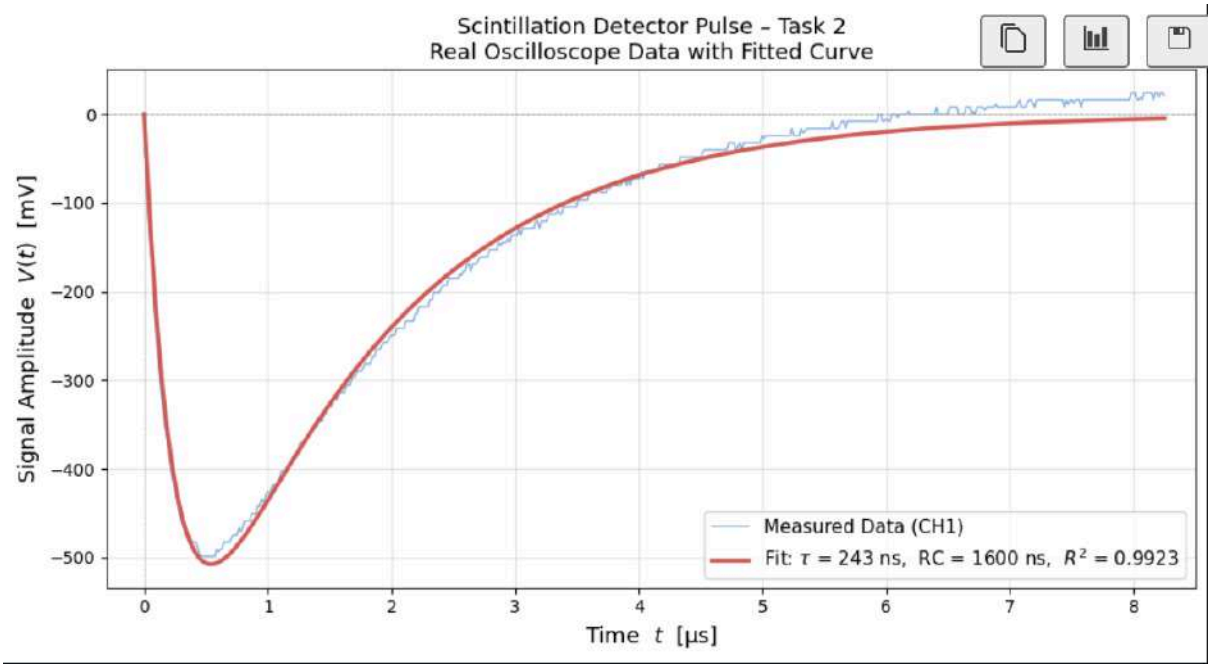


Figure 2: Figure of the simulated data points due and the fitting based on V(t) = K(e^{-t/RC} - e^{-t/\tau}, resulting in the fitted parameters.

Measured data info:	**Fitted Parameters:**
Pulse start : t = -0.275 µs	K = -0.8389 V
Peak value : -498.49 mV at t = 0.175 µs	RC = 1599.8 ± 7.6 ns (1.60 µs)
Rising edge : 450 ns	τ = 243.3 ± 2.3 ns
Data points : 1651	R^2 = 0.9923

Literature τ for NaI(Tl) ≈ 230 ns (deviation 5.8 %)

The curve fitting yielded a scintillation time constant tau ≈ 228ns and an electronic time constant RC ≈ 5651ns. The obtained value for tau is in excellent agreement with the accepted literature value for a NaI(Tl) crystal, which is approximately 230 ns. The RC time constant reflects the specific capacitance and resistance of our electronic circuitry.

Task 3: Amplitude of signal by Spectroscopy Amplifier (Vamp)

> **Task 3.** Determine the amplitude of the signal generated by the spectroscopy amplifier (V_{amp}) at different pulse shaping times and different gain settings.

In Task 3, we investigated the relationship between the spectroscopy amplifier's gain setting and its output signal amplitude Vamp. We adjusted the supply voltage to 500 V to produce an un-amplified detector signal with an amplitude of approximately 50 mV for the full energy peak.

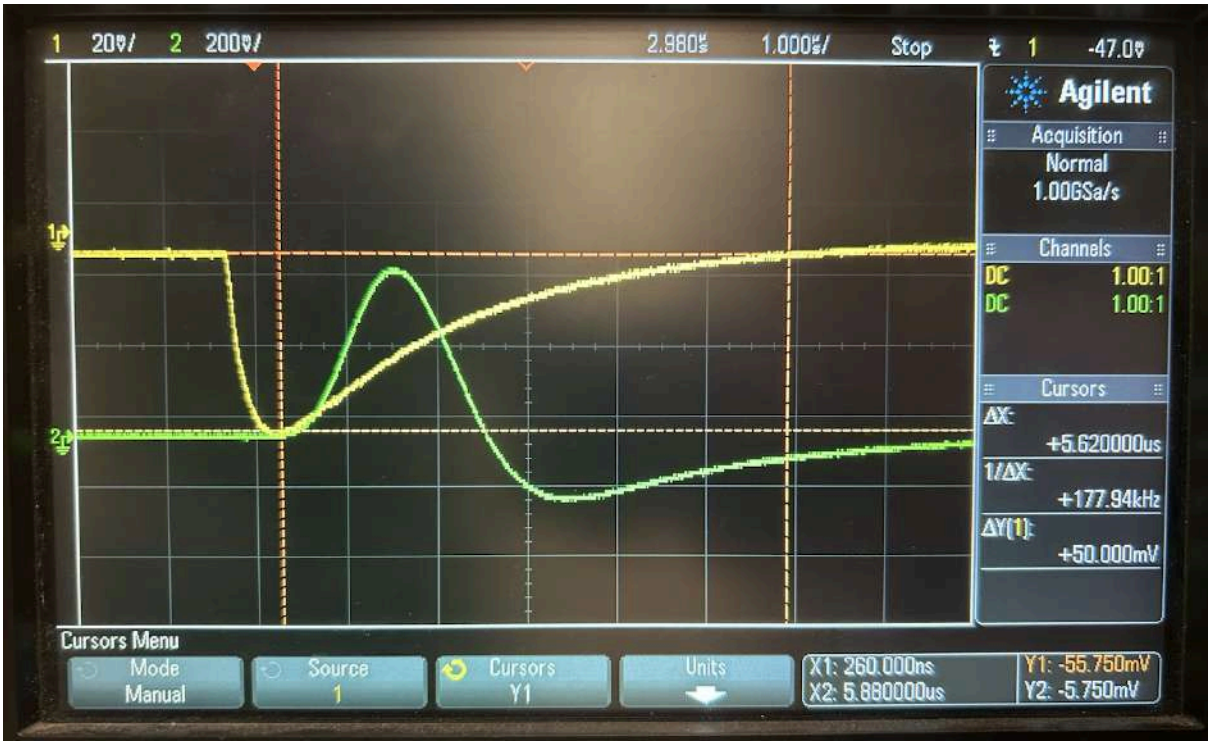


Image 3. A close-up of the signal wave at 500V applied voltage to form a 50mV signal amplitude

When we increase shaping time, the V_{amp} amplitude decreases and it increases in width. The highest signal is when the shaping time is 0.5 us.

> **Hint for the Lab Report.** Check by fitting, whether the amplitude of the amplifier output signal changes linearly with the gain. Calculate the actual amplification for the different settings. Document your observations!

We recorded the amplifier's output signal amplitude for various Coarse Gain settings while keeping the Fine Gain at unity. We then plotted these values and performed a linear regression to verify the linearity of the amplifier's response.

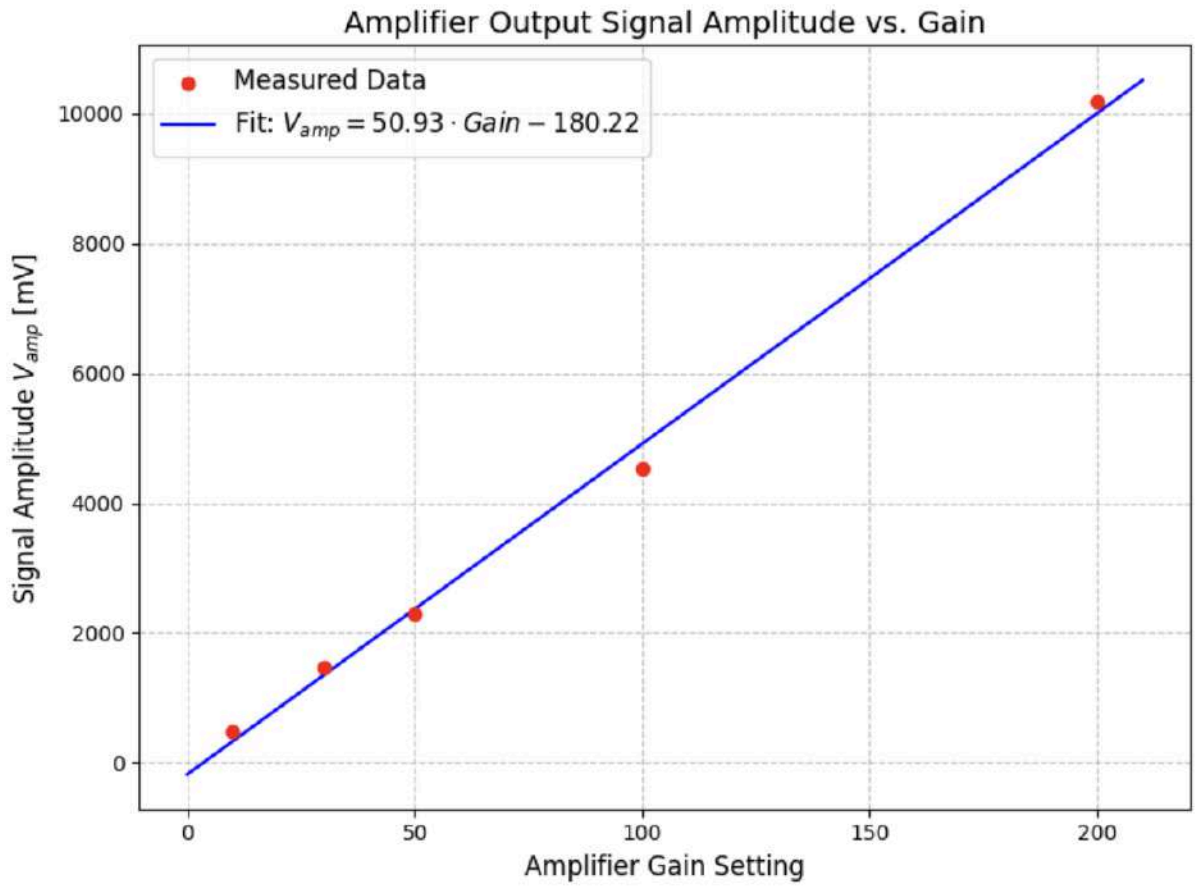


Figure 3: The plotted measured data of the amplifier course gain vs singal amplitude and the fitted line with the parameters:

--- Linear Fit Results ---

Slope (m): 50.93 mV/Gain

Intercept (c): -180.22 mV

R-squared: 0.9964

The linear fit of the data points yielded an R^2 value of 0.9964. This value is extremely close to 1, confirming that the output amplitude Vamp indeed scales linearly with the amplifier gain setting, as expected for a spectroscopy amplifier.

Task 4: Spectrum of MCA in Genie2000 Software

> **Task 4.** Observe the spectrum obtained by the multichannel analyzer in the Genie2000 software (see user manual in Sec. 3 of Part III).

For Task 4, we transitioned to using the multichannel analyzer (MCA) and Genie2000 software to observe the complete energy spectrum of the Cs-137 source. We set the applied high voltage to 720 V, the Coarse Gain to 10, and a Low Level Discriminator (LLD) of 0.1% to filter out low-energy electronic noise.

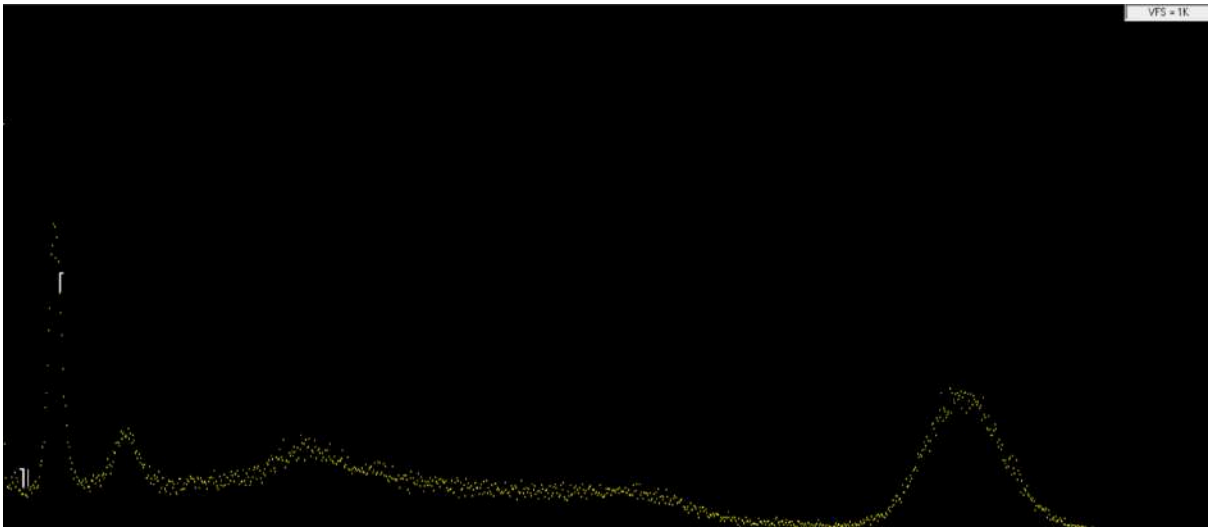


Image 4a: Result of the Genie2000 program

The recorded spectrum displays distinct features characteristic of gamma-ray interactions in the NaI(Tl) detector. The rightmost spike, which forms a Gaussian distribution, is the full energy peak (photopeak) representing gamma photons that deposited all their energy (661.7 keV) via the photoelectric effect. To its left, we can observe the Compton continuum and the Compton edge. The smaller peaks at the lower energy end are typically the backscatter peak and characteristic X-rays.

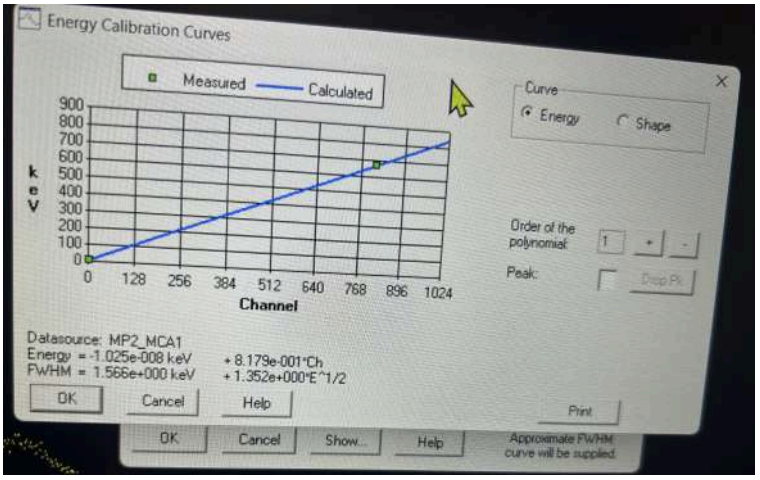


Image 4b: Energy calibration in the Genie2000 program, indicating energy per channel

Based on the software's energy calibration curve, the relationship between the energy and the channel number is given by the linear equation: Energy = -1.025e-08 keV + 0.8179 * Ch. From this, the channel-to-energy proportionality factor is determined to be exactly 0.8179 keV/channel.

Task 5: Measurements of the Cs sample in different environments

> **Task 5.** Identify and explain the peaks in the spectrum of the ^{137}Cs sample for the following cases.
>
> 1. Hold the detector in the air, and hold the source in front of it by tweezers.
> 2. Hold the source (by tweezers) close to the floor and the detector above it.
> 3. Detector and source are in the lead tower.

In Task 5, we compared the spectrum of the Cs-137 source in three different environmental setups to observe the effect of surrounding materials on the backscattered radiation. The baseline measurement, with both the detector and source enclosed in the lead tower, was already recorded in the previous task.

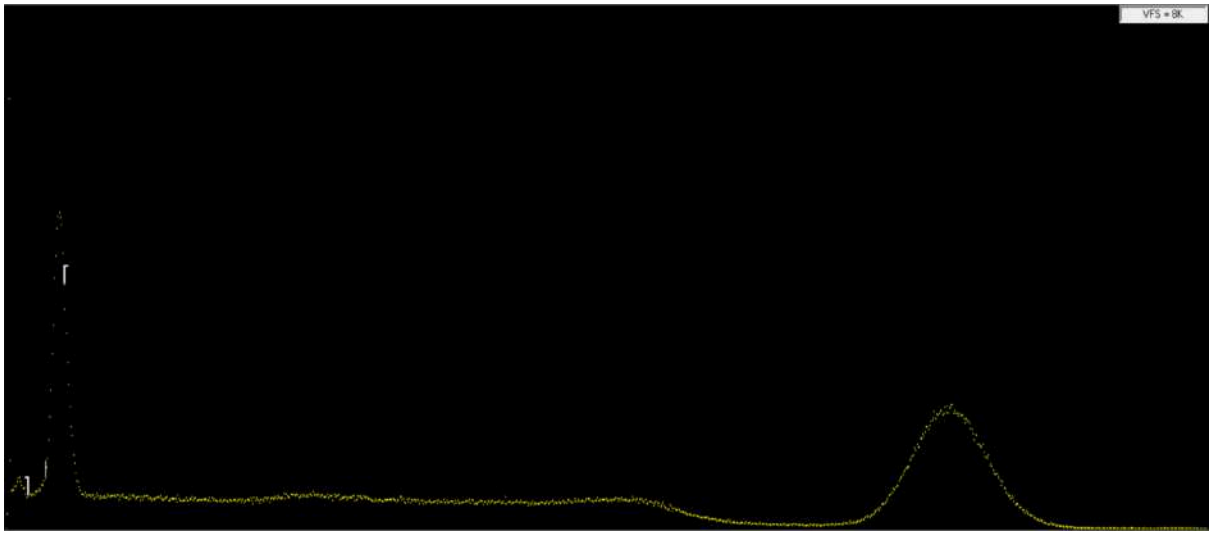


Figure 5a: The spectrum when the detector and Cesium are both held in the air

When both the detector and the Cs-137 source were held in the air, the spectrum notably changed. The prominent backscatter peak (the distinct spike normally seen at lower energies) essentially disappeared. This occurs because there was no dense shielding material nearby to scatter the gamma rays back into the detector.

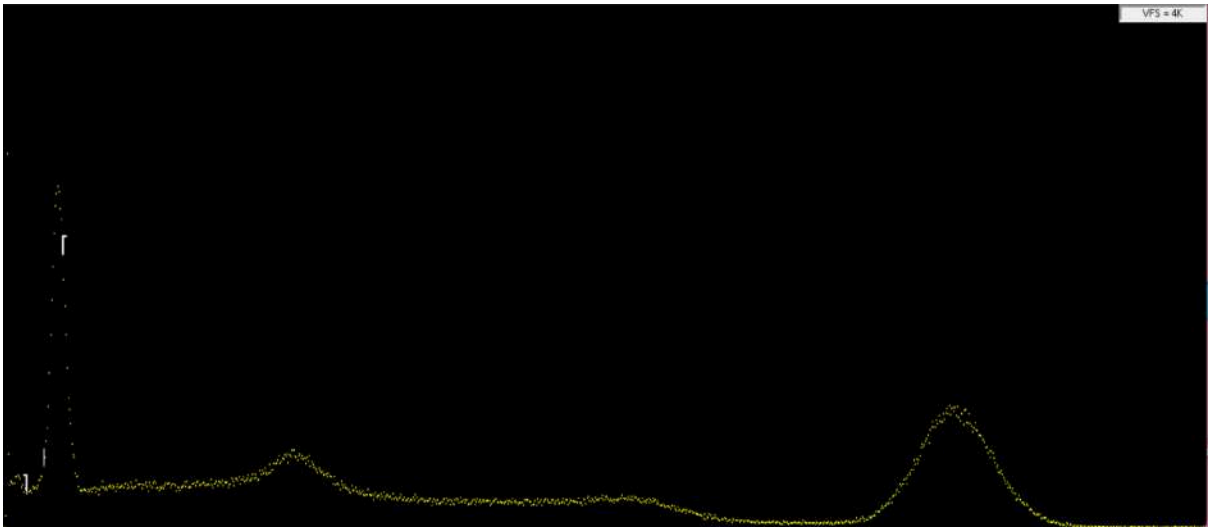


Figure 5b: The spectrum when the Cesium is on the floor, and the detector is on top of it

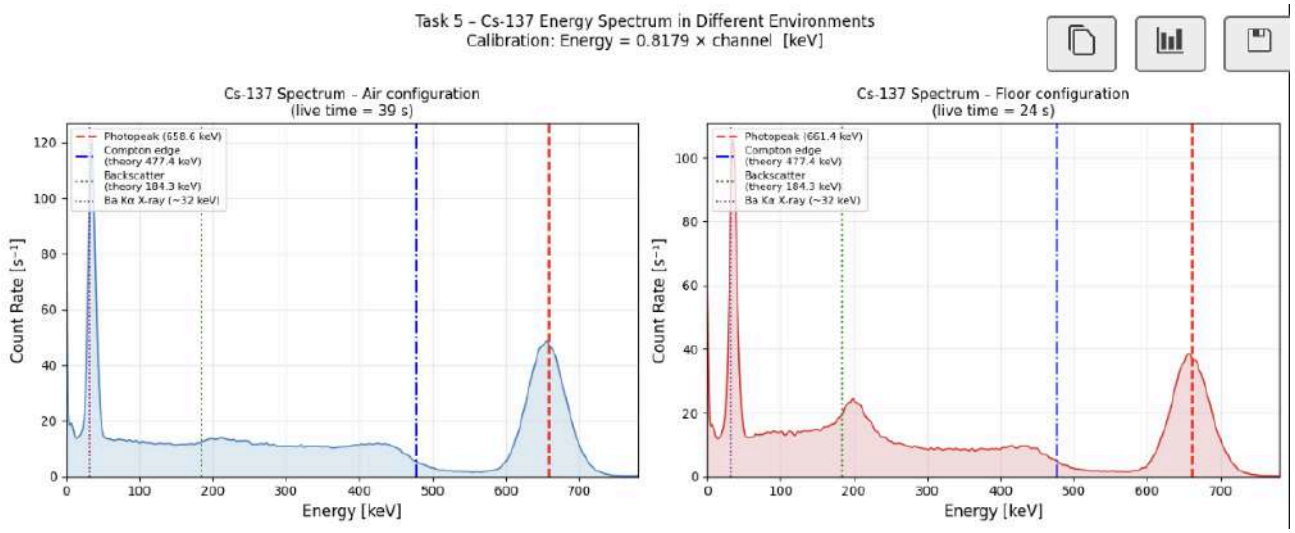
For the third configuration, the Cs-137 source was placed on the floor with the detector held directly above it. In this spectrum, the backscatter peak reappeared but was noticeably smaller in amplitude compared to when the setup was fully enclosed in the lead container.

This behavioral difference is due to the Compton backscattering of gamma rays from the surrounding surfaces. In the lead tower, the source is completely surrounded by high-density lead walls, maximizing the amount of radiation reflecting back into the detector. When placed on the floor, the gamma rays can only backscatter from the ground beneath the source, resulting in a reduced, but still measurable, backscatter peak. In the air, there is virtually no solid matter nearby to reflect the gamma rays, causing the peak to vanish entirely.

> **Hint for the Lab Report.** Calculate the channel-to-energy proportionality factor based on the recorded spectrum if energy calibration has not been performed during the measurement.
> Determine the energy of the peaks appearing in the spectrum.
> Calculate the expected energy of the Compton-edge (based on Eq. I.2.2), and compare it with the measured value.
> Discuss and document your observations and conclusions!

To perform a more detailed quantitative analysis, the spectra for the air and floor configurations were saved as .TKA files from Genie2000 and processed in Python. The energy calibration established in Task 4 (0.8179 keV/channel) was applied to convert channel numbers to energy values, and the count rates were obtained by dividing the raw counts by the live measurement time. The theoretical positions of the key spectral features were calculated from the

known Cs-137 photon energy of 661.7 keV using Eq. I.2.2 from the lab manual, and overlaid on the measured spectra for comparison.



Figures 5c, d: Energy spectrum in different environments

The theoretical positions of the key spectral features agree well with the measured values across both configurations. The photopeak falls within 0.5% of the known 661.7 keV value, the Compton edge at 468.7 keV is close to the theoretical 477.4 keV, and the backscatter peak at 198.7 keV is consistent with the theoretical 184.3 keV, with the small upward shift expected from contributions at non-180° scattering angles.

```
Theoretical Compton edge     : 477.37 keV  (ch 583.7)
Theoretical backscatter peak : 184.33 keV  (ch 225.4)

Measured photopeak (air)       : 658.6 keV
Measured photopeak (floor)     : 661.4 keV
Measured backscatter (floor)   : 198.7 keV
Measured Compton edge (air)    : 468.7 keV

Compton edge — theory vs meas  : 477.4 keV vs 468.7 keV  (diff = 8.7 keV)

── Peak Summary ──────────────────────────────────────────────────
Feature                      Theory [keV]  Measured [keV]   Config
------------------------------------------------------------------
Ba Kα X-ray                          32.0   ~35.2 (ch 43)     Both
Backscatter peak                    184.3           198.7    Floor
Compton edge                        477.4           468.7      Air
Photopeak (air)                     661.7           658.6      Air
Photopeak (floor)                   661.7           661.4    Floor
```

Image 5: Obtained data from the analysis of the spectrum

The measured photopeak energies confirm the energy calibration to within 0.5%. The Compton edge and backscatter peak both align closely with theory, with small deviations attributable to the finite energy resolution of the NaI(Tl) detector and the fact that backscattering in the floor configuration occurs across a range of angles rather than purely at 180°. The absence of the backscatter peak in the air configuration confirms that it originates entirely from Compton scattering in the surrounding dense material, with no contribution from the detector itself.

Section 2: Measurements with a neutron detector

Task 1: Neutron detector and spectrometry

Task 1. Set the operating point of the CHM-17/0485 type ^{3}He gas-filled proportional detector, where the

- signal-to-noise ratio is maximal,
- the sensitivity to the high voltage is minimal,
- and the complete spectrum is well distributed in the 0-10 V range of the MCA.

For the neutron detection section, the CHM-17/0485 type ^{3}He gas-filled proportional detector was connected to an in-house built preamplifier, a Canberra NIM 2022 spectroscopy amplifier, and the Canberra Multiport II MCA running Genie2000. The detector was positioned close to the Pu-Be neutron source. The high voltage was set to 1450 V, the coarse gain to 300, and the fine gain to 1024. By sweeping the shaping time from 0.5 μs upward and adjusting the LLD, we found that an LLD of 13.1% was the lowest setting at which the neutron peak was cleanly separated from the electronic noise floor.

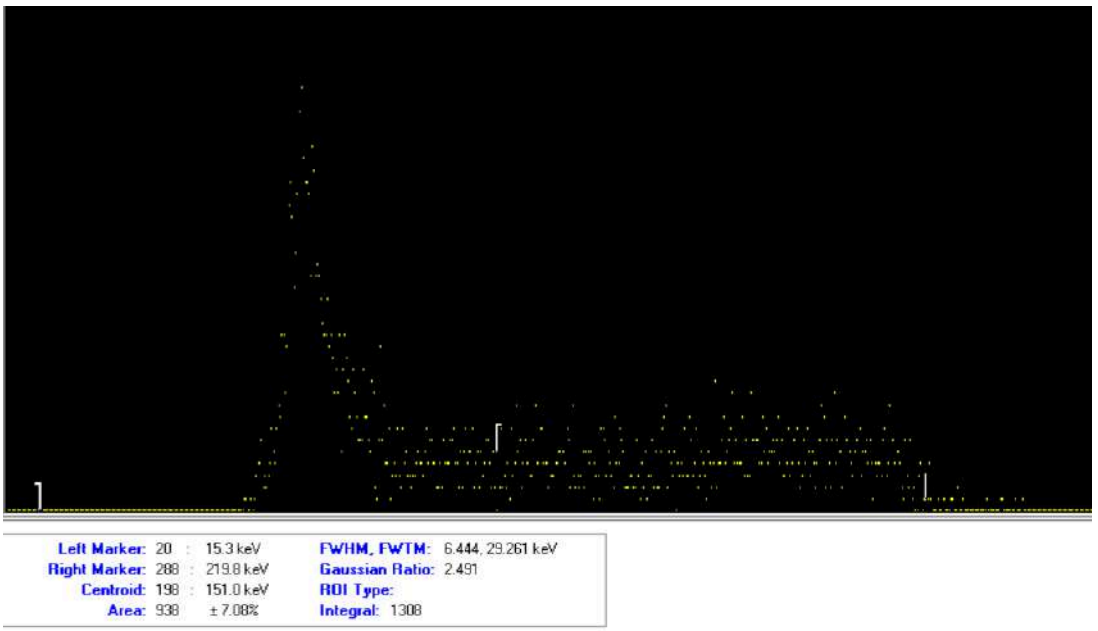


Figure 1.3a: Spectrum when the shaping time is 0.5us

At a shaping time of 0.5 µs the spectrum is poorly resolved, with a broad distribution centered at channel 198 (151 keV) and a small integral of 938 counts, indicating insufficient charge collection at this short shaping time.

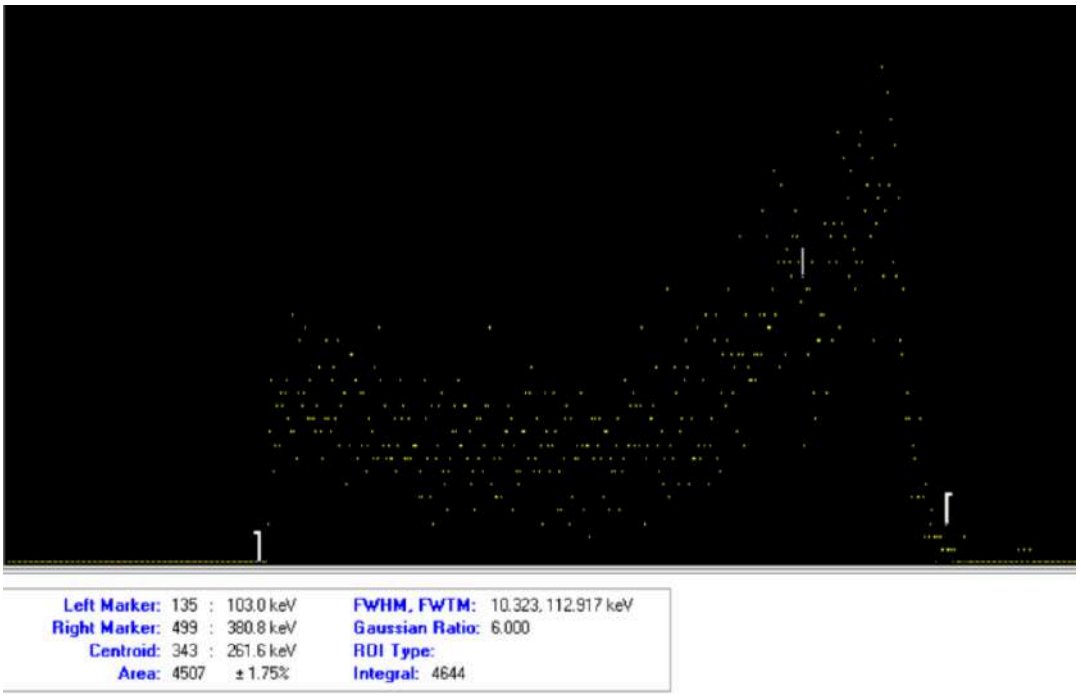


Figure 1.3b: Spectrum when the shaping time is 1us

Increasing the shaping time to 1 µs substantially improves resolution and statistics, with the centroid shifting to channel 343 (261.6 keV) and the integral rising to 4507 counts, reflecting more complete charge collection.

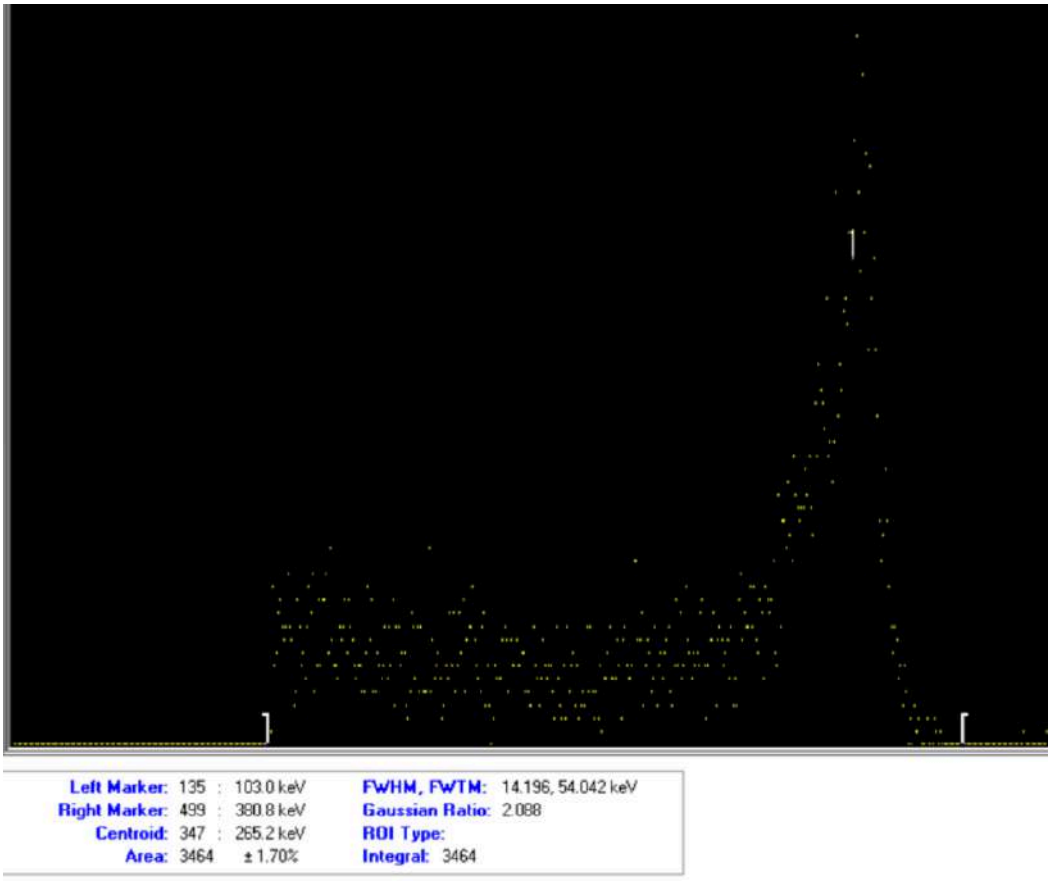


Figure 1.3c: Spectrum when the shaping time is 2us

A measurement at 2 µs confirms the improved resolution, with a centroid at channel 347 (265.2 keV) and an integral of 3464 counts, consistent with the previous result.

For the high-voltage scan, voltage was incremented in steps of 100 V rather than 50 V to reduce measurement time, with a shaping time of 2 µs and an LLD of 6.5%.

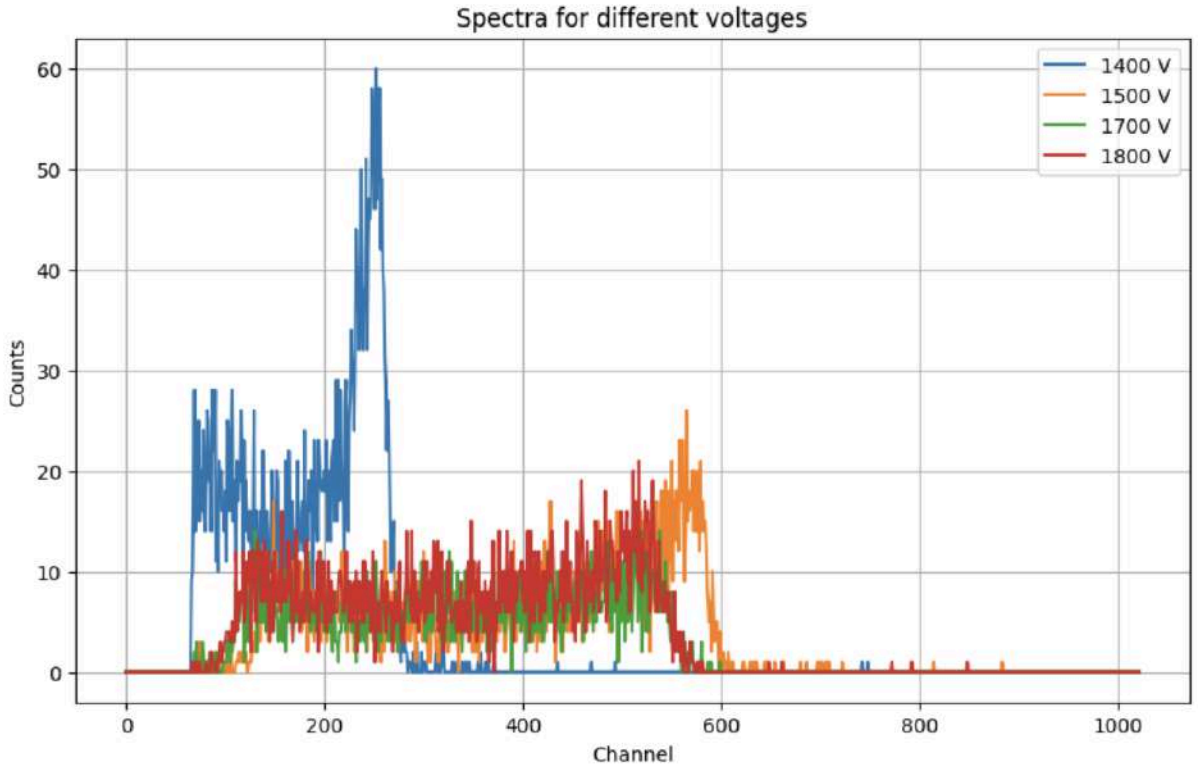


Figure 1.4: Combined spectra for different source voltages

Voltage (V)	1800	1700	1600	1500	1400	1300V
Amplification gain	50	50	130	240	240	
Integral (count)	3795	2910	3372	3665		
Measured time (s)	22.93	16.97	20.04	21.13		

Table 1.4: Note that that for 1300V it is no longer good because the the signal is now mainly electric noise instead of neutron detection will be excluded.

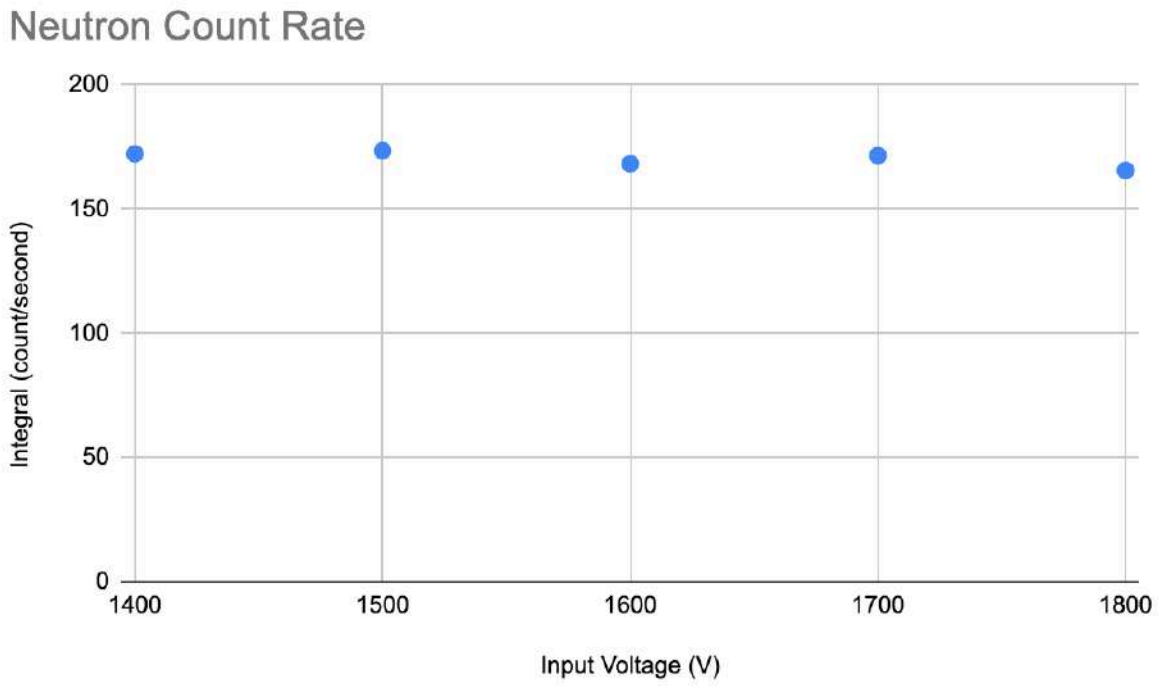


Figure 1.4: Calculating neutron count rate by dividing integral value by measured time (note that the y axis we called it integral when it should be neutron count rate, integral is just number of count)

The neutron count rate plot shows a stable plateau of approximately 165–170 counts/second across the 1400–1800 V range, confirming that 1600 V lies within the stable proportional region where the count rate is insensitive to small voltage fluctuations.

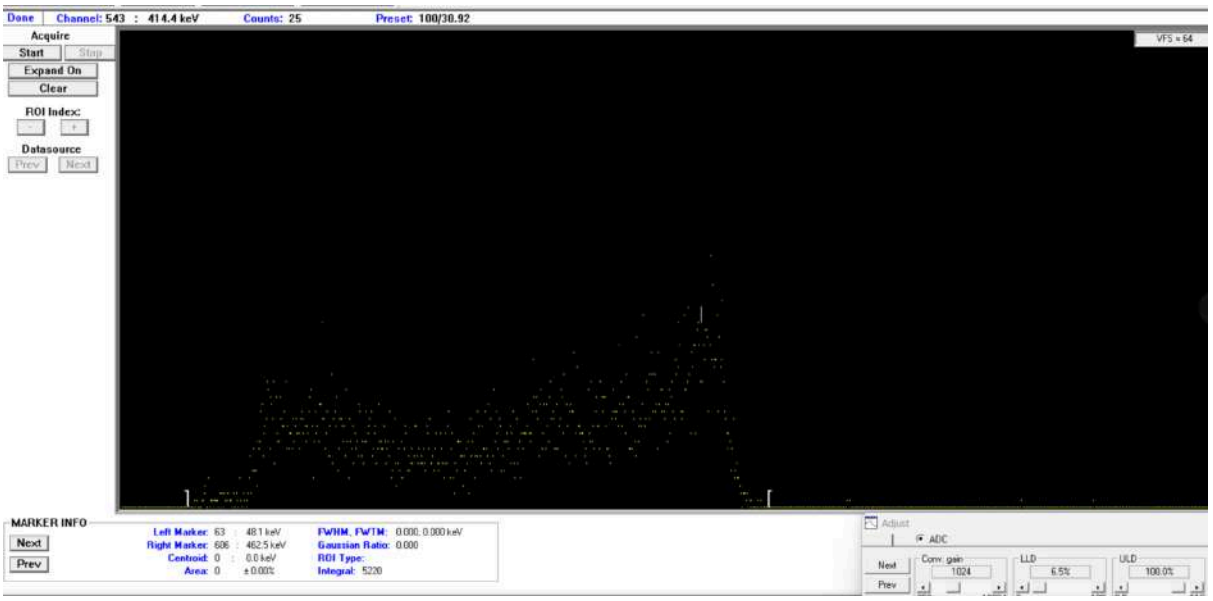


Figure 1.5. Spectrum of feasible operating point at 1600V source voltage and 115 amplification gain.

Task 2: Power supply - signal amplitude characteristics

Task 2. Determine the *power supply - signal amplitude* characteristics of the He-3 detector:

Hint for the Lab Report. Use the recorded spectrum data.
Find the channel corresponding to the full energy peak and convert its value (that is the channel number) to the amplitude of the un-amplified detector signal in $[V]$ units (using the nominal amplification values) for each of the power supply settings.
Plot the un-amplified detector signal in function of power supply voltages.
Document your observations.

For each high-voltage setting recorded in Task 1, the channel number of the full energy peak (764 keV) was identified in the Genie2000 spectrum and converted to the un-amplified detector signal amplitude using V = (peak channel / 1024) × 10 V / nominal gain.

The un-amplified signal rises steeply between 1400 V and 1700 V before levelling off, consistent with the onset and saturation of the proportional amplification region. The operating point of 1600 V was selected in the rising but stable portion of this curve.

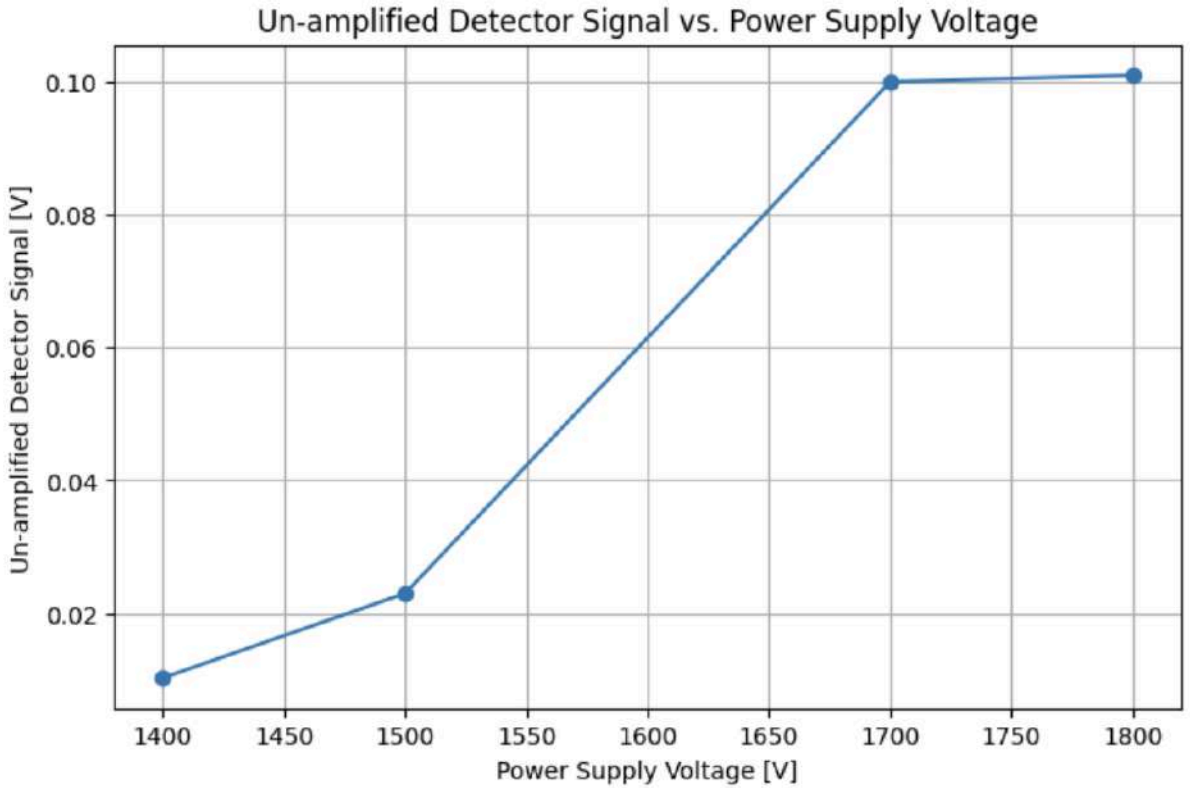


Figure 2: Un-amplified Detector Signal vs. Power Supply Voltage

Task 3: Set the SCA and the counter units to their operating point

Task 3. Set the SCA and the counter units to their operating point:

With the detector operating at 1600 V and a coarse gain of 115, the SCA was configured with an LLD of 15% to exclude electronic noise while retaining all neutron events. A 60 s measurement yielded an SCA counter reading of 9218 counts, compared to a Genie MCA integral of 9032 counts over the same live time — a difference of 2.1%, attributable to minor threshold alignment differences between the two systems.

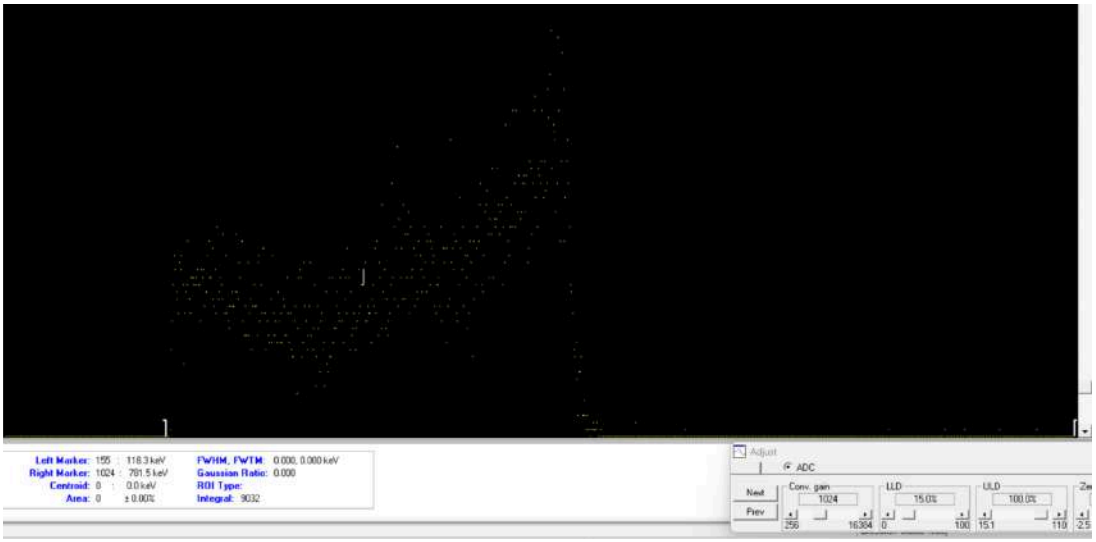


Figure 3: Spectrum operation gat 1600V, 115 amplitude gain, 15% LLD

Optional Task. Analyse the output signal of the SCA unit on the oscilloscope. Measure the width of the TTL pulse.

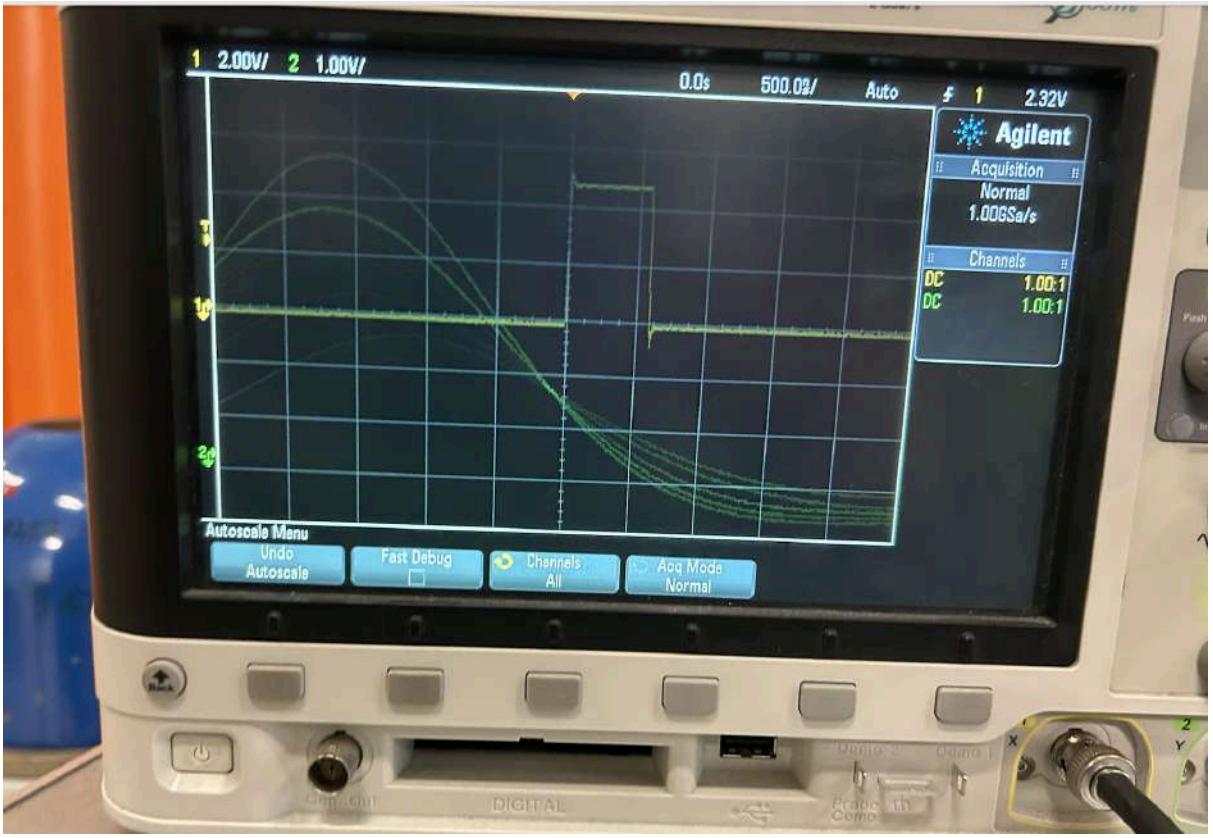


Image 3: Image showing the TTL pulse

The SCA output signal observed on the oscilloscope is a standard positive TTL pulse with a measured width of approximately 580 ns.

Task 4. Neutron count rate at different reactor powers up

Task 4. Measure the neutron count rate in the beam from the horizontal channel at different reactor powers up to max. 100 W and observe the dead time effect. This part has to be performed by all groups at the same time with the measurement setup completed in the previous task. Discuss the required reactor power levels with the lab supervisor, who will communicate with the operators.

With the detector and SCA set to their operating points, the neutron count rate from the reactor's horizontal irradiation channel was measured at increasing reactor power levels up to 20 W. Both the SCA-counter and Genie MCA systems recorded simultaneous 60 s measurements at each power level. At low power the count rate is strictly proportional to reactor power; at higher power the measured rate falls below the linear expectation due to dead time losses.

Both systems follow the linear extrapolation closely up to ~5 W, after which the measured count rates increasingly deviate below the expected linear trend, demonstrating the onset of dead time losses.

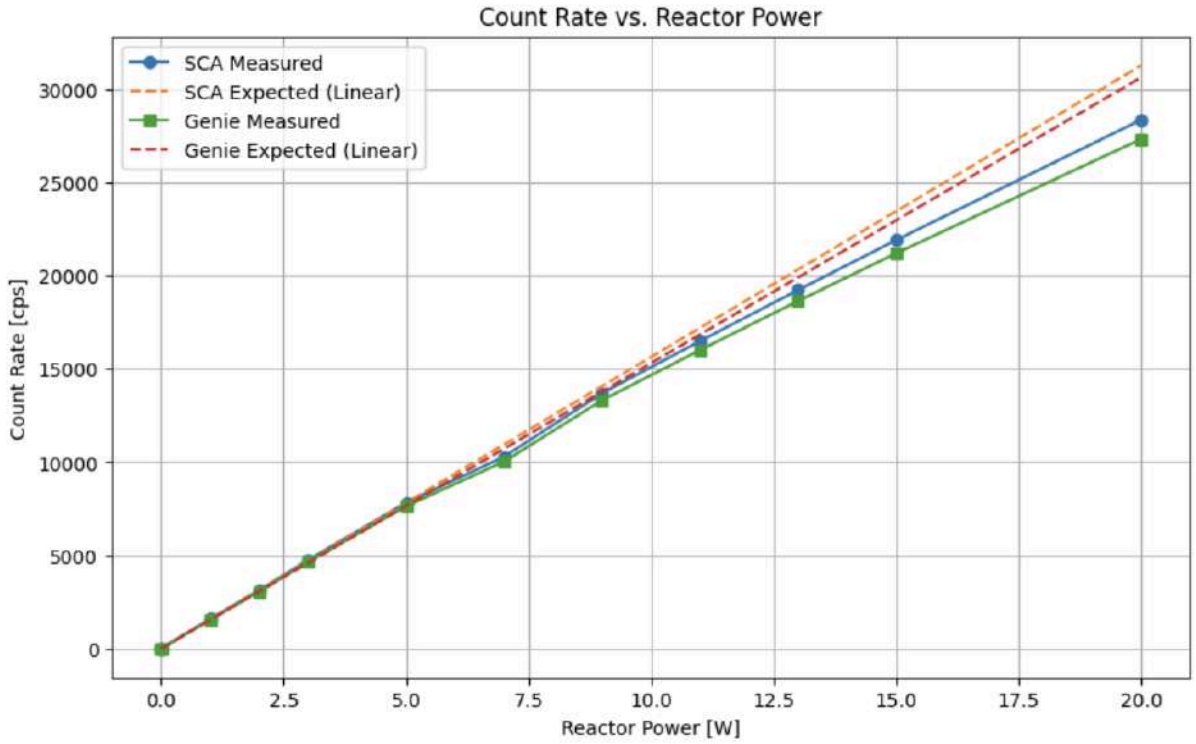


Figure 4: Figure of count rate vs reactor power from SCA and Genie2000, with the expected count as an extension of 0-2W range

At 20 W the SCA system loses 9.31% of counts and the Genie MCA 10.77%, consistent with the MCA incurring greater processing overhead. The proportionality factors derived from the low-power linear region are 1560 cps/W (SCA) and 1527 cps/W (Genie). Using Eq. I.4.2, the dead times are estimated at 3.2 μs (SCA) and 3.8 μs (Genie) — both substantially larger than the 580 ns TTL pulse width, confirming that the dominant dead time originates in the internal electronic processing stages rather than the output logic pulse.

Proportionality Factor: The proportionality factor (k) relates the reactor power (P) to the true count rate (n). We derived this factor from the low-power measurements (1W to 2W), where dead time effects are negligible (approximately 0 percent).

SCA Proportionality Factor: 1560 cps per Watt

Genie Proportionality Factor: 1527 cps per Watt

Dead Time Estimation: Using the data measured at the highest reactor power (20W), we calculated the system dead time (tau) based on Equation 1.4.2 from the lab manual: tau = (n - m) / (n * m), where n is the expected true count rate and m is the measured count rate.

SCA System at 20W: n = 31200 cps m = 28366 cps Dead time (tau) = 3.2 microseconds

Genie MCA System at 20W: n = 30540 cps m = 27330 cps Dead time (tau) = 3.8 microseconds

Comparison with TTL Pulse Width The measured electronic dead time (3.2 to 3.8 microseconds) significantly exceeds the SCA TTL output pulse width of 580 nanoseconds. This confirms that the system dead time is not constrained by the duration of the output TTL pulse itself, but rather by the cumulative processing time of the internal electronic stages, including pulse shaping, analog-to-digital conversion, and software processing.

Conclusion

This exercise demonstrated the full nuclear measurement chain from detector to spectrum analysis across two complementary detection systems. For the NaI(Tl) scintillation detector, the PMT gain exponent was found to be 7.04, consistent with the theoretical range of 6–9. Fitting the real oscilloscope waveform yielded a scintillation time constant of 243 ns, in good agreement with the NaI(Tl) literature value of 230 ns, and an electronic time constant RC of 1600 ns. The spectroscopy amplifier was confirmed to be highly linear ($R^2 = 0.9964$), and the Cs-137 energy spectrum was successfully calibrated and analysed, with all major spectral features — photopeak, Compton edge, and backscatter peak — identified and compared against theoretical predictions. For the ^{3}He neutron detector, the operating point was established at 1600 V with a stable count rate of ~165–170 cps. Dead time analysis at 20 W reactor power yielded system dead times of 3.2 µs (SCA) and 3.8 µs (Genie MCA), both significantly exceeding the 580 ns TTL pulse width and confirming that the bottleneck lies in internal electronic processing rather than the output logic.